

Test Strategy

Our test strategy was centered on unit testing using the JUnit framework, which integrates seamlessly with Maven. The primary goal was to validate each component of the application in isolation before verifying their integration. This bottom up approach ensures that individual pieces of logic are correct before they are combined, simplifying debugging and increasing confidence in the system as a whole. Our strategy was implemented in two main layers.

Model Unit Testing:

- Objective: To verify the core "business logic" of the application.
- Execution: Each class in the model package (e.g., Student, Course, Session) was tested independently.
- Example: CourseTest validates that the logic for enrollment, waitlisting, and capacity limits functions correctly without needing any user input or file system.

Application Integration Testing:

- Objective: To verify the interaction between the user interface logic (Menu), the application state (Session), and external dependencies (like the Transaction class for file I/O).
- Execution: MenuTest simulates user interaction by providing a mock Scanner with pre-defined inputs.
- Example: The testAddStudentUI test provides fake user input (ID, name, email) and asserts that the Session object is correctly modified. The testSaveAndLoadUI test verifies that the Menu class can correctly trigger file system operations and that the data is

persisted and retrieved accurately, ensuring the test cleans up any generated files afterward.

By testing both the isolated logic (unit tests) and the component interactions (integration tests), we were able to build a comprehensive test suite that validates the application's correctness from the core model up to the user facing application layer.

TERMINAL OUTPUT:

```
[INFO] -----
```

```
[INFO] T E S T S
```

```
[INFO] -----
```

```
[INFO] Running edu.uca.registration.app.MenuTest
```

```
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.224 s -- in  
edu.uca.registration.app.MenuTest
```

```
[INFO] Running edu.uca.registration.model.CourseTest
```

```
Roster full, adding to waitlist
```

```
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in  
edu.uca.registration.model.CourseTest
```

```
[INFO] Running edu.uca.registration.model.EnrollmentTest
```

```
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.020 s -- in  
edu.uca.registration.model.EnrollmentTest
```

```
[INFO] Running edu.uca.registration.model.SessionTest
```

[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -- in
edu.uca.registration.model.SessionTest

[INFO] Running edu.uca.registration.model.StudentTest

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in
edu.uca.registration.model.StudentTest

[INFO] Running edu.uca.registration.repo.ConfigManagerTest

[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s -- in
edu.uca.registration.repo.ConfigManagerTest

[INFO]

[INFO] Results:

[INFO]

[INFO] Tests run: 24, Failures: 0, Errors: 0, Skipped: 0

[INFO]

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 2.383 s

[INFO] Finished at: 2025-11-01T20:54:25-05:00

[INFO] -----